

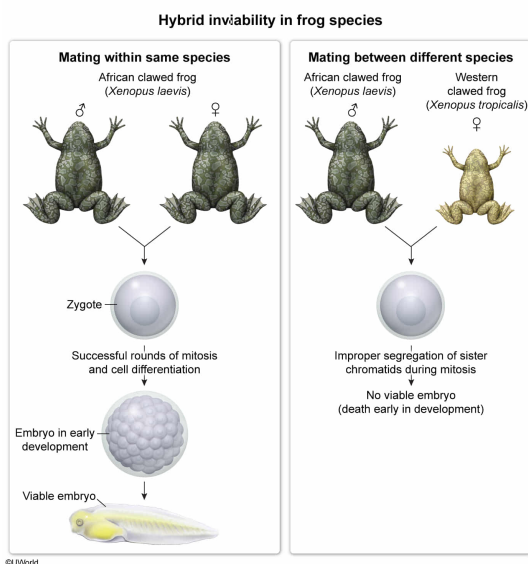
A male African clawed frog (*Xenopus laevis*) and a female Western clawed frog (*Xenopus tropicalis*) mate and produce multiple hybrid embryos that die early in development. Given this, the deaths of the hybrid embryos are most likely caused by which of the following?

- ✔ ☒ A. Improper segregation of sister chromatids during embryonic mitosis [41%]
- ☐ B. Sperm and egg incompatibility [21%]
- ☐ C. Reduced production of gametes in the parental species [1%]
- ☐ D. Improper pairing of homologous chromosomes during meiosis in the embryo [35%]

Correct

42% answered correctly

Explanation:



Hybrid offspring are produced when members of two different species interbreed. The mating of hybrid offspring with members of either parental species may cause **genetic leakage**, or the transfer of genetic information *between* species. However, in some cases two species may be unable to produce hybrid embryos, or hybrids may be inviable (unable to survive) or infertile. The inability of two species to produce viable or fertile hybrid offspring is called **reproductive isolation**, which may be **prezygotic** (before **zygote formation**) or **postzygotic** (after zygote formation).

In this scenario, the mating of a male African clawed frog (*Xenopus laevis*) and a female Western clawed frog (*Xenopus tropicalis*) **produces hybrid embryos** that die early in development—an example of *postzygotic* reproductive isolation. **Mitosis** is essential for embryonic growth and production of the organism's somatic cells. Failure of sister chromatids to segregate properly (eg, due to **chromosomal nondisjunction**) **during embryonic mitosis** generates cells containing *missing* or *extra* genetic information. Accordingly, the ability of these cells to divide and differentiate into functional somatic cells in the embryo would be *impaired*. This would inhibit zygote development and likely cause early embryonic death.

(Choice B) Sperm and egg incompatibility in the two species would *prevent* fertilization and the formation of an embryo. Because the mating of these frog species *produced* embryos, the sperm and egg must be *compatible*.

(Choice C) Gametes (eg, eggs, sperm) form via **meiosis** during gametogenesis. *Reduced* sperm and egg production in parental species would *decrease* the probability of fertilization and zygote formation following a mating event. However, the frogs are described as having produced *multiple* hybrid embryos, and reduced gamete formation in the parental species would *not* affect the development of any embryos formed after mating.

(Choice D) Meiosis is the process by which diploid organisms produce haploid gametes. Although the *parents* needed to perform proper meiosis to produce the offspring embryos, errors by the *offspring embryos* would not be apparent until the offspring attempt to sexually reproduce. The question states that the embryos died during development, before they had a chance to attempt sexual reproduction.

Educational objective:

Hybrid offspring are produced from the mating of two different species. However, reproductive isolation (prezygotic or postzygotic) can prevent different species from producing viable or fertile hybrid offspring.

Time spent:0 sec

Subject:
Biology

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System:
Genetics and Evolution